

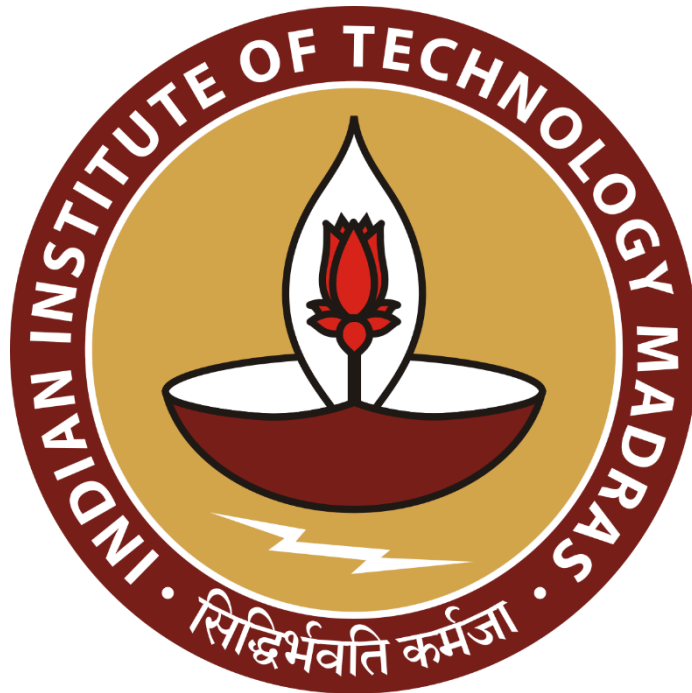
Analysis of Sales Trends and Inventory Management in a Women's Ethnic Wear Business

A Mid-Term report for the BDM capstone Project

Submitted by

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1 Executive Summary

Tulsi is a women's ethnic wear shop located in New Market, Bhopal. The store mainly sells kurta sets, readymade suits and unstitched fabrics. In daily operations, the shop faces some challenges related to managing inventory and handling changing customer demand. Sales are not the same throughout the year and usually increase during wedding seasons. Apart from this, most of the sales data is written manually in a diary, so detailed product-wise information is not available. Because of this, it becomes difficult to clearly understand which items are selling more and which are not.

The data for this study was collected directly from the store using primary sources. Purchase data was taken from physical bills provided by vendors, while sales data was recorded from the store's daily diary, where only total daily revenue is maintained. The dataset includes details such as date, total sales amount, tax values, product category, quantity purchased and purchase cost. Since item-wise sales data is not available, the analysis is based on total values. Basic descriptive methods like monthly totals and comparisons were used to understand how sales and purchases changed over the three months. It was observed that sales were much higher in January, mainly due to wedding season demand.

The analysis was done using Microsoft Excel and Python. Simple methods like trend analysis, comparison of sales and purchase values, and inventory turnover were used. The results indicate that the business maintains around a 50% profit margin but faces issues when demand suddenly drops after peak season. This creates extra stock in some months. Overall, the study indicates that better use of data can help the store plan inventory more effectively and maintain stable performance.

2 Proof of Data Originality

Details: The business uses two primary methods for recording data:

1. Purchase Records: Maintained through physical bills provided by vendors, which include details such as product category, quantity, and purchase cost.
2. Sales Diary: A manual diary is used to record daily total revenue.

Raw Data : 📁 raw data

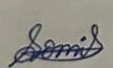
- Shop Name: Tulsi
- Type of Organisation: B2C (Business-to-Consumer)
- Shop's Address: New Market, Bhopal, M.P. (462003)
- Shop Owner's Name: Somil jain
- Owner's Phone Number: 7389840404
- Number of Employees: 5
- Working Time: 11am - 9.30pm

Photo Proof: 📷 photo proof



Video Proof:  Tulsi Video.mp4

Authorization letter signed by the owner:

	
मध्य तल, 17- अ, न्यू मार्केट, टी.टी. नगर, भोपाल, फोन : 0755-2552685	
TIN - 23974100980	दिनांक. 25-03-2026
Date: 25 March, 2026	
<u>TO WHOM IT MAY CONCERN</u>	
This letter is to formally authorize Mr. Naman Gupta, a student of the IIT Madras BS Degree Program, to undertake a Business Data Management Capstone Project on behalf of our business. We hereby grant him permission to access and utilize our business data, including sales and purchase records, for the purpose of conducting this academic project. The objective of this project is to analyze our business operations and derive meaningful insights that can help improve inventory management, sales performance, and overall decision-making. Mr. Naman Gupta has been entrusted with the responsibility of analyzing and managing the data to generate actionable insights and recommendations that may contribute to the enhancement of our business processes. We confirm that the data shared is genuine and has been provided with our full consent solely for academic purposes. We appreciate his efforts and support him in successfully completing this project.	
	
Name: SOMIL JAIN Designation: Owner / Proprietor Business Name: Tulsi (Women's Ethnic Wear Store) Location: New Market, Bhopal Contact Number: 7389840404	

3 Metadata

Datasheet: [+ 23f2002076](#)

Collab File: [BDM_project.ipynb](#)

Data collection duration :3 months

Data collection dates: 01-12-2025 to 28-02-2026

NOTE: The data for this study was collected in raw form from the store, where purchase records were maintained through physical vendor bills and later organized into Excel format. In contrast, sales data was recorded manually in a diary, where only total daily sales were noted without item-wise details. Due to this, the dataset combines detailed purchase information with aggregated daily sales values, which was used for further analysis.

Table 1: Daily Sales Data

Dataset Dimensions: The dataset contains 8 columns and 79 rows



S.NO	MONTH	DATE	CASH MEMO	AMOUNT	CGST 2.5%	SGST 2.5%	TOTAL SALES
1	December 2025	2.12.25	5818-5820	13210	330	330	13870
2	December 2025	3.12.25	5821-5824	17800	445	445	18690
3	December 2025	4.12.25	5825-5827	8701	217	217	9135
4	December 2025	5.12.25	5828-5829	3524	88	88	3700
5	December 2025	6.12.25	5830	6666	167	167	7000
6	December 2025	7.12.25	5831-5835	16557	414	414	17385
7	December 2025	9.12.25	5836	6666	167	167	7000
8	December 2025	10.12.25	5837-5839	12615	315	315	13245
9	December 2025	11.12.25	5840-5841	5896	147	147	6190
10	December 2025	12.12.25	5842-5843	10081	252	252	10585
11	December 2025	13.12.25	5844-5846	26119	653	653	27425
12	December 2025	14.12.25	5847-5849	17096	427	427	17950
13	December 2025	16.12.25	5850-5852	17185	430	430	18045
14	December 2025	17.12.25	5853-5854	10428	261	261	10950
15	December 2025	18.12.25	5855-5856	13139	328	328	13795

Fig 3.1: A snapshot of how the data looks like (top 15 entries)

- **S.NO:** Serial number for each sales entry.
- **MONTH:** The month in which the sale was recorded.
- **DATE:** Specific date of the transaction.
- **CASH MEMO:** Bill or receipt number for the sale.
- **AMOUNT:** Base value of sales before tax.
- **CGST:** Central Goods and Services Tax applied on the sale.
- **SGST:** State Goods and Services Tax applied on the sale.
- **TOTAL:** Final total sales amount including taxes.

Table 2: Purchase Data

Dataset Dimensions: The dataset contains 9 columns and 190 rows

Tulsi - Purchase Data (December 2025 - February 2026)								
Estimated selling price has been calculated using an approximate profit margin of 50% as stated by the store owner and is used only for analytical purposes.								
S.NO	MONTH	ARTICLE CODE	CATEGORY	QUANTITY	PURCHASE PRICE	TOTAL PURCHASE	ESTIMATE SELLING PRICE	Estimated Sales Value
1	December 2025	KAF20426	KURTI SET	8	1444	11552	2166	17328
2	December 2025	NI20256	KURTI SET	10	1045	10450	1568	15680
3	December 2025	TAPS17603	KURTI SET	8	1832	14656	2748	21984
4	December 2025	RRF7860	KURTI SET	18	1045	18810	1568	28224
5	December 2025	CB612	KURTI SET	5	1045	5225	1568	7840
6	December 2025	CB642	KURTI SET	10	1045	10450	1568	15680
7	December 2025	RRF7860	KURTI SET	18	1045	18810	1568	28224
8	December 2025	RRF7814	KURTI SET	12	1045	12540	1568	18816
9	December 2025	GRC20571	KURTI SET	15	1045	15675	1568	23520
10	December 2025	KAF20426	KURTI SET	8	1444	11552	2166	17328
11	December 2025	SAF400163	KURTI SET	10	1045	10450	1568	15680
12	December 2025	GOUR7332	KURTI SET	5	1045	5225	1568	7840
13	December 2025	FAB1697	KURTI SET	12	1045	12540	1568	18816
14	December 2025	TAPS17607	KURTI SET	16	1832	29312	2748	43968
15	December 2025	SRMR79113	KURTI SET	10	1045	10450	1568	15680

Fig 3.2: A snapshot of how the data looks like (top 15 entries)

- **S.NO:** Serial number for each purchase entry.
- **MONTH:** Month in which the purchase was made.
- **ARTICLE:** Code or identifier of the product.
- **CATEGORY:** Type of product (e.g., Kurti Set, Suit, Fabric).
- **QUANTITY:** Number of units purchased.
- **PURCHASE PRICE:** Cost price per unit of the product.
- **TOTAL PURCHASE:** Total cost of purchased items (quantity $\times$ price).
- **SELLING PRICE:** Estimated selling price per unit based on margin.
- **ESTIMATED SALES:** Estimated total revenue from the purchased items.

Table 3: Monthly Sales vs Purchase Summary

B. SALES vs PURCHASE — MONTH-WISE COMPARISON			
Month	Trading Days	Actual Sales (Rs)	Purchase Spend (Rs)
December 2025	26	313132	311440
January 2026	28	758606	407633
February 2026	25	309129	461796

Fig 3.3: A snapshot of the monthly sales and purchase summary data

- **Month** : Represents the month for which data is recorded
- **Trading Days** : Number of days store was open for sales
- **Actual Sales** : Total revenue generated from sales during the month
- **Purchase Spend** : Total amount spent on purchasing inventory in month

4 Descriptive Statistics

The descriptive statistical analysis was performed using the cleaned sales and purchase datasets in Python (Google Colab).

1. Descriptive Statistics of Daily Sales Data (Fig 4.1)

sales.describe()					
	SNO	AMOUNT	CGST	SGST	TOTAL
count	79.00	79.00	79.00	79.00	79.00
mean	40.00	16646.95	416.19	416.19	17479.30
std	22.95	11738.63	293.47	293.47	12325.60
min	1.00	1048.00	27.00	27.00	1100.00
25%	20.50	6666.00	167.00	167.00	7000.00
50%	40.00	15377.00	384.00	384.00	16145.00
75%	59.50	22981.00	574.50	574.50	24130.00
max	79.00	54757.00	1369.00	1369.00	57495.00

Fig 4.1

- The average daily sales come out to be around ₹16,646, while the median is about ₹15,377, which shows that most of the days have sales close to this range without extreme imbalance.
- The sales vary quite a lot, with the minimum going as low as ₹1,048 and the maximum reaching ₹54,757. This clearly shows that some days have very low sales while some days perform exceptionally well.
- The standard deviation is relatively high (around ₹11,738), which indicates that daily sales are not consistent and fluctuate significantly, reflecting changing customer demand across different days.

2. Descriptive Statistics of Monthly Sales Trend (Fig 4.2)

```

days_per_month = sales.groupby("MONTH")["DATE"].count()
monthly_sales = sales.groupby("MONTH")["TOTAL"].sum()

avg_daily_sales = monthly_sales / days_per_month

# Overall average (across all months)
overall_avg_sales = sales["TOTAL"].mean()

print(avg_daily_sales)
print("Overall Average Sales:", overall_avg_sales)

... MONTH
Dec    12043.46
Feb    12365.16
Jan    27093.07
dtype: float64
Overall Average Sales: 17479.303797468354

```

(Fig 4.2)

- On average, the store makes around ₹17,479 per day over the three months.
- January clearly stands out, with an average daily sale of about ₹27,093. This is much higher than the other months and is mainly because of the wedding season, when demand for ethnic wear increases.
- December and February are quite similar, with average sales of around ₹12,043 and ₹12,365 respectively. These months reflect more normal business conditions, and the drop after January shows how sales slow down once the peak season is over.

3. Descriptive Statistics of Sales vs Purchase Comparison (Fig 4.3)

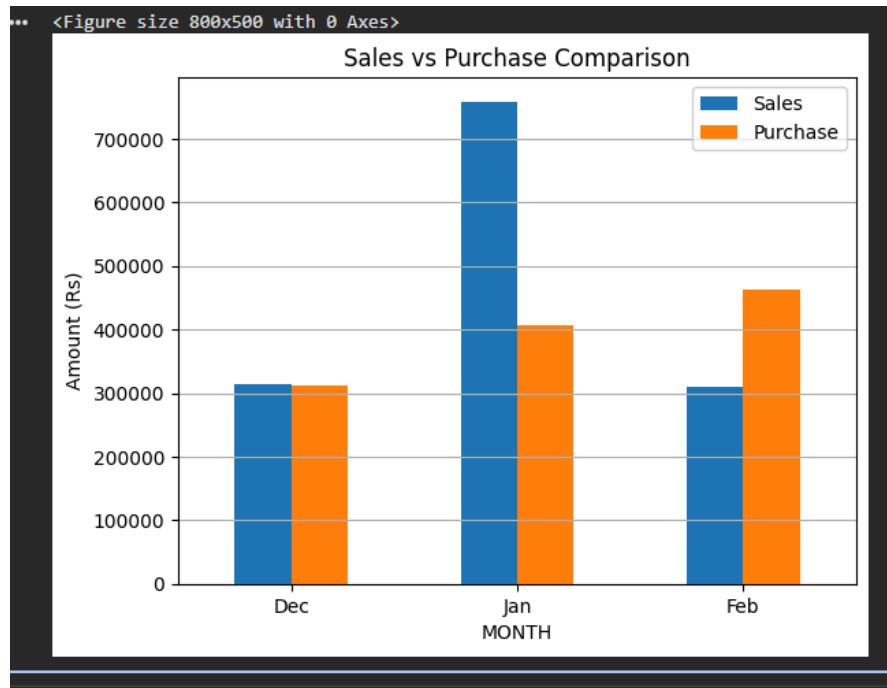


Fig 4.3

- The comparison between sales and purchase shows that January has the highest sales, which is well supported by higher purchases, indicating strong demand during that period.
- In February, the purchase spend is noticeably higher than the actual sales, which suggests that the store bought more inventory than required, leading to excess stock.
- December shows a relatively balanced relationship between sales and purchase, reflecting stable demand and better alignment between inventory and sales.

4. Descriptive Statistics of Category-wise Purchase Data (Fig 4.4)

`purchase.describe()`

	SNO	QUANTITY	PURCHASE_PRICE	TOTAL_PURCHASE	SELLING_PRICE	ESTIMATED_SALES	PROFIT
count	190.00	190.00	190.00	190.00	190.00	190.00	190.00
mean	95.50	6.17	1048.65	6215.10	1573.25	9324.74	3109.64
std	54.99	7.77	276.92	5898.59	415.32	8850.36	2951.77
min	1.00	1.00	415.00	835.00	623.00	1253.00	418.00
25%	48.25	3.00	907.00	2506.25	1361.00	3760.75	1254.50
50%	95.50	4.00	1045.00	4326.00	1568.00	6490.00	2164.00
75%	142.75	8.00	1106.50	8360.00	1659.75	12544.00	4184.00
max	190.00	77.00	2204.00	31955.00	3306.00	47971.00	16016.00

`purchase.groupby("CATEGORY")[["PURCHASE_PRICE", "TOTAL_PURCHASE"]].describe()`

CATEGORY	PURCHASE_PRICE								TOTAL_PURCHASE							
	count	mean	std	min	25%	50%	75%	max	count	mean	std	min	25%	50%	75%	max
KURTI SET	63.00	1134.90	232.24	893.00	1045.00	1045.00	1105.00	1994.00	63.00	9423.95	6238.99	2414.00	4432.00	7976.00	12046.00	29312.00
READYMADE SUIT	13.00	1401.31	507.73	1045.00	1045.00	1045.00	1994.00	2204.00	13.00	11858.38	7013.34	4180.00	7976.00	10450.00	15675.00	30800.00
SINGLE KURTI	65.00	908.52	227.63	415.00	787.00	903.00	1045.00	1360.00	65.00	5415.89	4662.65	2075.00	3340.00	4180.00	5456.00	31955.00
UNSTITCHED FABRIC	49.00	1030.06	168.35	730.00	893.00	997.00	1150.00	1570.00	49.00	1652.41	1086.45	835.00	997.00	1150.00	1880.00	5860.00

- The category **Single Kurti** has the highest number of purchase entries (65 SKUs), indicating that the store stocks more variety in this category, likely to meet regular day-to-day demand.
- In terms of pricing, **Readymade Suit** shows the highest average purchase price of around ₹1,401 per unit, suggesting that it consists of relatively premium products compared to other categories.
- The total purchase value is highest for **Kurtti Set** (approximately ₹9,424 on average per entry), indicating that this category receives the largest investment and plays a major role in the store's inventory.

5 Detailed Explanation of Analysis Process/Method

5.1 Data Cleaning and Preprocessing

The data was initially collected in an unstructured format, where sales were recorded manually in a diary and purchase data were available through physical bills. The first step involved converting this data into Excel sheets. While doing this task, care was taken to ensure that all entries were correctly transferred without duplication or missing values.

In Excel, basic cleaning was performed, such as removing blank rows, aligning columns, and checking totals using simple formulas. After importing the data into Python (Google Colab), further cleaning was done using pandas, where column names were standardised and unnecessary columns were removed. Date values were also formatted properly to allow month-wise analysis.

This step was important because even small errors in data (like wrong totals or missing values) could affect the final analysis. Cleaning ensured that the dataset used for analysis was accurate and consistent.

5.2 Analysis Process

The analysis was carried out step by step using a combination of Excel and Python libraries such as pandas and numpy:

1. Data Import: The cleaned Excel file was uploaded to Google Colab and read using pandas (pd.read_excel), which allowed easy handling of tabular data.
2. Data Verification: Basic checks such as total sales and purchase values were cross-verified using Excel totals and pandas sum() functions to ensure consistency.
3. Descriptive Statistics: The describe() function in pandas was used to calculate statistical measures such as mean, median, minimum, maximum, and standard deviation for both sales and purchase data.
4. Monthly Aggregation: Sales data was grouped by month using groupby() to calculate total sales and number of trading days. Average daily sales were then calculated using:
Average Daily Sales = Total Monthly Sales / Number of Days
5. Comparative Analysis: Sales and purchase values were compared month-wise using grouped data to identify whether inventory was aligned with actual sales.
6. Category-wise Analysis: Purchase data was grouped by category to analyze differences in number of SKUs, purchase price, and total investment across product types.
7. Profit Estimation: Since item-wise selling price was not available, estimated values were calculated using:

$$\text{Estimated Selling Price} = \text{Purchase Price} \times 1.5$$

$$\text{Estimated Profit} = \text{Estimated Sales} - \text{Total Purchase}$$

Note: This was based on the 50% margin provided by the store owner.

8. Visualisation: Graphs such as line charts (for trends) and bar charts (for comparisons) were created using Matplotlib to better understand patterns in the data.

5.3 Justification of Methods

The approach used in this analysis was selected according to the type of data available. Since detailed item-level sales information was not accessible, it was more appropriate to rely on descriptive statistical techniques rather than complex analytical models. Measures such as mean and standard deviation were used to get a clear understanding of the overall sales pattern and the variation within it.

To observe trends over time, the data was aggregated on a monthly basis. This helped in identifying seasonal patterns, particularly the effect of the wedding season on sales performance. In addition, comparing sales data with purchase values made it possible to detect situations where excess stock may have been maintained.

A category-wise breakdown of the data further enhanced the analysis by showing how inventory is distributed among different product segments. For implementation, tools like pandas and numpy were used for efficient data processing, while Excel supported initial data organization and validation. Altogether, these methods provided a simple yet effective way to analyze and interpret the data.

6 Results and Findings

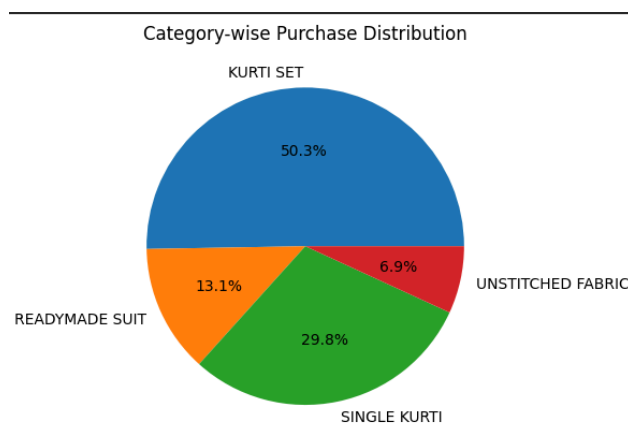
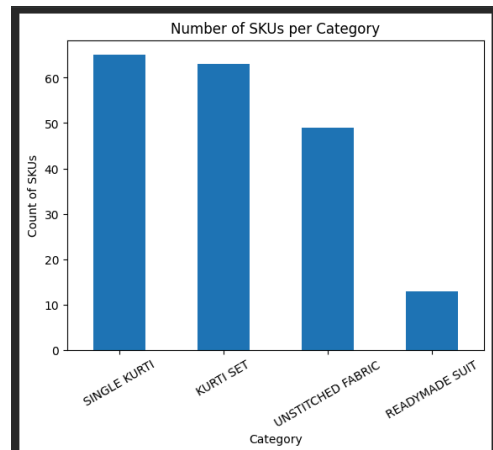


Fig 6.1

Kurti Set alone accounts for around **50.3%** of total purchase value, showing that the store heavily invests in this category and depends on it for major sales. Single Kurti contributes nearly **29.8%**, indicating strong focus on variety-based products, while Readymade Suit (13.1%) represents a smaller but higher-value segment. Unstitched Fabric has the lowest share (**6.9%**), suggesting limited inventory allocation, but this also means lower risk compared to heavily stocked categories.



Single Kurti has the highest number of SKUs (65), showing that the store maintains the most variety in this category to meet regular customer demand. Kurti Set (63 SKUs) is also heavily stocked, indicating that it is a key category for the store in terms of product range. Other categories like Unstitched Fabric (49 SKUs) and Readymade Suit (13 SKUs) have comparatively fewer SKUs, suggesting limited variety and lower focus.

Findings:

- The analysis clearly shows a strong **seasonal impact**, with sales peaking in January due to the wedding season and dropping in February. This indicates that the store's performance is highly dependent on specific high-demand periods.
- Inventory is heavily concentrated in **Kurti Set and Single Kurti**, both in terms of investment and number of SKUs, showing that the store focuses more on these categories to drive sales.
- **Readymade Suit**, despite having fewer SKUs, has the highest average purchase price, indicating that it represents a **premium segment**. However, its lower volume suggests limited demand compared to other categories.
- The current pattern also shows a risk of **overstocking**, especially when purchases remain high even after demand declines. Better alignment between seasonal demand and purchasing decisions can help improve inventory efficiency.